

Project Summary: Enterprise Network Design

Enterprise Network Design is a Cisco Packet Tracer project that demonstrates the design and configuration of a secure multi-site business network connecting an HQ location and a Branch location. The network uses VLAN segmentation, 802.1Q trunking, OSPF dynamic routing, and extended ACLs to provide controlled communication between different departments and sites.

The HQ network is divided into three VLANs: **ADMIN VLAN 10**, **IT VLAN 20**, and **GUEST VLAN 30**. The Branch network uses a separate subnet to represent a remote office location. OSPF is configured between the HQ and Branch routers to automatically exchange routes and support inter-site connectivity. Switch trunking is used to carry multiple VLANs across the HQ switch uplink.

Security is enforced using an extended access control list on the HQ router. The ACL blocks the Guest VLAN from accessing the Admin and IT networks while still allowing Guest users to communicate with the Branch network. This simulates a realistic enterprise security policy where guest users are isolated from internal corporate resources but can still access approved network destinations.

The project was validated using Cisco IOS commands and end-to-end testing. Verification included checking VLAN assignments, trunk status, OSPF neighbor relationships, routing tables, ACL hit counts, and ping results. Successful tests confirmed that Branch connectivity works, while restricted Guest-to-Admin access is properly blocked.

This project demonstrates practical networking skills in enterprise LAN/WAN design, VLAN planning, routing, access control, troubleshooting, and documentation.

Key Technologies

- Cisco Packet Tracer
- VLANs
- 802.1Q Trunking
- OSPF Routing
- Extended ACLs
- Cisco IOS CLI
- Network troubleshooting and validation

Key Features

- Multi-site HQ and Branch topology
- VLAN segmentation for Admin, IT, and Guest users
- OSPF dynamic routing between routers
- Trunk configuration for VLAN traffic

- Guest network isolation using ACL rules
- Successful Branch connectivity validation
- Blocked Guest access to protected Admin/IT networks
- Verification using routing tables, OSPF neighbors, VLAN tables, trunk status, ACL hit counts, and ping tests